

Memory Management in Mobile Applications

Benefits of Efficient Memory Usage

Memory Constraints Influence Data Structure & Algorithm Selection

Examples of Algorithms with Different Space Complexities

Algorithm Efficiency

Limited Device Memory

- Choose In-place Algorithms
- $O(1)$ Space
- Avoid Large Temporary Arrays & Deep Recursion
- Prefer Compact Data Structures
- Arrays, Hashing

Memory Usage

Amount of RAM used during execution

Space Complexity

Total memory needed as input increases

Auxiliary Space

Extra temporary memory (excluding input data)

Input Size

Number of elements or records (n)